
PROFILE

- **Research-Driven Applied AI:** Delivering multimodal foundation models, LLM/VLM fine-tuning, and RAG systems. Published at NeurIPS, KDD, IJCAI, WACV, and CVPR Workshop combining rigorous research with production-ready systems.

RESEARCH INTERESTS

Multimodal learning and vision-language models — LLM/VLM fine-tuning, preference optimization, and knowledge distillation — Retrieval-augmented generation (RAG) and structured reasoning.

EDUCATION

University of Virginia (UVA)*Ph.D. in Computer Science, CGPA: 3.97/4.00*

Advisor: Madhav Marathe (Multimodal AI, Vision-Language Models)

Aug 2022 – May 2027

Charlottesville, VA

EXPERIENCE

AI/ML Intern – GenAI (LIHA)*LinkedIn*

Summer 2026

Sunnyvale, CA

- **LLM Fine-Tuning:** Optimized hiring assistant using knowledge distillation, preference learning, and DPO-inspired objectives for aligned, knowledge-grounded responses.
- **Preference Modeling:** Modeled user choice signals to improve candidate ranking accuracy and recruiter productivity at scale.

Graduate Research Assistant*Biocomplexity Institute, UVA*

Nov 2022 – Present

Charlottesville, VA

- **Multimodal AI & VLMs:** Designed vision-language pipelines using foundation models (CLIP, RemoteCLIP, SAM, YOLO, DINOv2) for large-scale geospatial reasoning.
- **Distributed ML Systems:** Trained LLM/VLM models on 1M+ samples using PyTorch DDP, FSDP, LoRA/QLoRA, and diffusion-based augmentation.
- **Retrieval-Augmented Generation:** Built RAG pipelines combining dense retrieval with multimodal reasoning for factual, explainable predictions.

SELECTED PUBLICATIONS

IRRISIGHT (NeurIPS 2025)

Large-scale multimodal dataset (1.4M patches); vision-language benchmark for irrigation classification; 58% improvement from multimodal fusion.

IrrMap (KDD 2025)

National-scale dataset and benchmark suite for irrigation mapping; evaluated segmentation models across geospatial, visual, and text modalities.

KIIM (IJCAI 2025)

Knowledge-informed multimodal classification; fused domain priors with satellite data; +51% improvement over baselines.

PRISM-CAFO (WACV 2026)

Prior-conditioned remote-sensing pipeline for infrastructure detection; DINOv2 + SAM2 integration; 90%+ accuracy on CAFO mapping.

AG-STELLA (IJCAI 2026)

Spatio-temporal transformer for crop and livestock monitoring; +16% macro accuracy; +27% F1 improvement.

CAFOSat (CVPR Workshop 2026)

Diffusion-based, prompt-guided synthetic augmentation with GroundingDINO + Stable Diffusion inpainting for CAFO infrastructure; synthetic achieves 0.764 F1 (only 0.6% below 0.770 F1 real data).

NOTABLE PROJECTS

LIHA – LinkedIn Hiring Assistant: Applied LLM fine-tuning with knowledge distillation and preference optimization to improve candidate ranking and recruiter matching.

Visual Scene Grounding with Knowledge Graphs: Structured knowledge priors supporting grounded generation; knowledge graphs feeding retrieval-augmented reasoning for multimodal understanding.

Knowledge Graph Creation from Remote Sensing: Ongoing: Building structured knowledge from satellite imagery using RAG for information retrieval and question answering over geospatial scenes.

SKILLS

Core AI/ML: LLM/VLM fine-tuning · Preference learning (DPO, RLHF) · Knowledge distillation · Retrieval-augmented generation · Multimodal learning · Computer vision

Foundation Models & Fine-tuning: Prithvi (Transformer) · Qwen · CLIP/RemoteCLIP · SAM · YOLO · DINOv2 · LoRA/QLoRA · SFT

Agentic AI & Knowledge Systems: LangChain · LangGraph · Flyte · Redis · MongoDB · RAG orchestration

Systems & Scale: PyTorch · PyTorch Lightning · Hugging Face · DeepSpeed · FSDP/DDP · CUDA · Distributed training

Languages & Tools: Python · Java · C/C++ · Git · Bash · Docker · Linux

LEADERSHIP, MENTORSHIP & PROFESSIONAL SERVICE

- Reviewer, NeurIPS 2025 and KDD 2025 – Top-tier venues for machine learning and data mining research.
- Mentor to undergraduate researchers on multimodal AI, LLM fine-tuning, and reproducible ML practices.
- Lead organizer, UVA Computer Science Research Symposium 2025.

FELLOWSHIPS, HONORS & AWARDS

- Ph.D. Fellowship, Department of Computer Science, University of Virginia.
- Travel Grants, NeurIPS 2025 and IJCAI 2025.
- Ranked 1st in section merit list, Bangladesh University of Engineering and Technology (BUET).